

Questions about critical care and emergency nursing theory

Acid and base imbalance and ABG interpretation

- 1. What is the main function of the buffer system in the body?**
 - A. Increase blood pressure
 - B. Neutralize bacteria
 - C. **Regulate acid-base balance**
 - D. Produce energy

- 2. Which of the following is one way the kidneys help maintain acid-base balance?**
 - A. Excrete bicarbonate into urine
 - B. **Reabsorb bicarbonate from urine**
 - C. Produce hydrogen ions
 - D. Store carbon dioxide

- 3. Which organ compensates more quickly for acid-base imbalance?**
 - A. **Kidneys**
 - B. Liver
 - C. Lungs
 - D. Pancreas

- 4. A patient has the following arterial blood gases: PaCO₂ 33, HCO₃ 15, pH 7.23. Which condition below is presenting?**
 - A. Metabolic alkalosis partially compensated
 - B. **Metabolic acidosis partially compensated**
 - C. Respiratory alkalosis not compensated
 - D. Metabolic acidosis fully compensated

5. A patient with COPD has the following blood gases: PCO₂ 59, pH 7.26, HCO₃ 42. Which of the following conditions is presenting?

A. Respiratory alkalosis

B. Respiratory acidosis

C. Metabolic alkalosis

D. Metabolic acidosis

Triage

7. What is the triage category for a patient who needs immediate life-saving intervention?

A. Non-urgent

B. Urgent

C. Emergent

D. Delayed

8. Which healthcare professional typically performs triage in the ED?

A. Pharmacist

B. Lab Technician

C. Registered Nurse

D. Radiologist

9. What is the *first and most critical* goal of an effective triage system?

A. Minimize documentation

B. Identify life-threatening conditions

C. Discharge stable patients early

D. Perform diagnostic tests

10. A patient experiencing cardiac arrest falls under which triage category?

A. Nonurgent

B. Urgent

C. Emergent

D. Delayed

Acute Respiratory Distress Syndrome (ARDS)

1. **What is the main cause of Acute Respiratory Distress Syndrome (ARDS)?**
 - A) Bacterial infection in the brain
 - B) **Severe fluid buildup in the lungs**
 - C) Low blood sugar
 - D) Joint inflammation
2. **What is the primary treatment for ARDS?**
 - A) Antibiotics only
 - B) **Respiratory support with mechanical ventilation**
 - C) Heart surgery
 - D) Physical therapy
3. **What clinical consequence results from impaired gas exchange in ARDS?**
 - A) Hyperglycemia
 - B) **Hypoxia**
 - C) Bradycardia
 - D) Hypercalcemia
4. **What long-term complication can develop as a result of this ongoing lung injury?**
 - A) Pulmonary embolism
 - B) **Lung fibrosis**
 - C) Asthma
 - D) Pleural effusion
5. **Why should nurses avoid overexerting ARDS patients during care?**
 - A. It increases risk of bleeding
 - B. It can cause dehydration
 - C. **It increases oxygen demand and fatigue**
 - D. It raises blood sugar levels

Pulmonary embolism

1. **What is the most common source of thrombotic emboli in pulmonary embolism?**
 - A. Pulmonary veins
 - B. **Deep veins of the legs**
 - C. Left atrium
 - D. Coronary arteries

2. A massive PE typically occludes what percentage of the pulmonary vascular bed?
 - A. Less than 20%
 - B. 25–30%
 - C. More than 40%
 - D. 100%
3. Which of the following cardiac chambers is primarily affected by the hemodynamic changes in PE?
 - A. Left atrium
 - B. Right ventricle
 - C. Left ventricle
 - D. Aorta
4. What is the major hemodynamic consequence of a significant pulmonary embolism?
 - A. Pulmonary edema
 - B. Pulmonary hypertension
 - C. Aortic dissection
 - D. Left ventricular hypertrophy

Shock questions

1. Shock is best defined as:

- A. Decrease in blood pressure
- B. Inadequate tissue perfusion leading to cellular dysfunction
- C. Failure of the heart to pump blood
- D. Decrease in blood volume

Answer: B

2. The main cause of tissue hypoxia in shock is:

- A. Increased oxygen consumption
- B. Imbalance between oxygen supply and demand
- C. Decrease in red blood cell count
- D. Excessive vasodilation

Answer: B

3. The hallmark of all types of shock is:

- A. Hypotension
- B. Ineffective tissue perfusion
- C. Tachycardia
- D. Metabolic acidosis

Answer: B

4. Hypovolemic shock results from:

- A. Myocardial infarction
- B. Loss of blood or plasma volume
- C. Widespread vasodilation
- D. Severe allergic reaction

Answer: B

5. Cardiogenic shock occurs due to:

- A. Loss of sympathetic tone
- B. Decreased intravascular volume
- C. Inability of the heart to pump effectively
- D. Bacterial infection

Answer: C

6. Septic shock is a type of:

- A. Cardiogenic shock
- B. Hypovolemic shock
- C. Distributive shock
- D. Obstructive shock

Answer: C

7. Neurogenic shock results from:

- A. Massive blood loss
- B. Loss of sympathetic nervous system tone
- C. Hypersensitivity reaction
- D. Infection

Answer: B

8. Which of the following is not a distributive shock?

- A. Septic shock
- B. Neurogenic shock
- C. Anaphylactic shock
- D. Cardiogenic shock

Answer: D

9. In the initial stage of shock, cells switch from:

- A. Aerobic to anaerobic metabolism
- B. Anaerobic to aerobic metabolism
- C. Protein to fat metabolism
- D. Glucose to lipid metabolism

Answer: A

10. The major byproduct of anaerobic metabolism is:

- A. Pyruvic acid
- B. Lactic acid
- C. Citric acid
- D. Amino acids

Answer: B

11. During the compensatory stage, the sympathetic nervous system causes:

- A. Vasodilation
- B. Decreased heart rate
- C. Vasoconstriction and increased heart rate
- D. Increased urine output

Answer: C

12. Activation of the renin-angiotensin-aldosterone system (RAAS) during shock causes:

- A. Vasodilation
- B. Sodium and water retention
- C. Decreased blood volume
- D. Decreased blood pressure

Answer: B

13. During the progressive stage of shock, failure of the sodium-potassium pump leads to:

- A. Cellular dehydration
- B. Cellular swelling and dysfunction
- C. Increased ATP production
- D. Improved perfusion

Answer: B

14. The refractory stage of shock is characterized by:

- A. Reversible cellular injury
- B. Effective compensatory mechanisms
- C. Irreversible organ failure and MODS
- D. Increased cardiac output

Answer: C

15. A mean arterial pressure (MAP) less than ____ mmHg indicates inadequate tissue perfusion.

- A. 100
- B. 80
- C. 60
- D. 50

Answer: C

16. A patient in shock has restlessness, confusion, and agitation. These signs indicate:

- A. Adequate cerebral perfusion
- B. Early cerebral hypoxia
- C. Severe dehydration
- D. Respiratory alkalosis

Answer: B

17. Decreased urine output in shock is due to:

- A. Decreased renal perfusion
- B. Increased ADH
- C. Increased renal blood flow
- D. Excessive fluid therapy

Answer: A

18. Which laboratory finding is expected in progressive shock?

- A. Decreased lactic acid
- B. Increased liver enzymes
- C. Increased urine output
- D. Normal BUN and creatinine

Answer: B

19. Disseminated Intravascular Coagulation (DIC) may develop during:

- A. Initial stage
- B. Compensatory stage
- C. Progressive stage
- D. Recovery stage

Answer: C

20. The first priority in managing shock is:

- A. Administer IV fluids
- B. Maintain airway and oxygenation
- C. Give vasopressors
- D. Monitor urine output

Answer: B

21. Fluid resuscitation in hypovolemic shock is best initiated with:

- A. Blood transfusion
- B. Colloid solution
- C. Crystalloid solution (e.g., normal saline)
- D. Dextrose solution

Answer: C

22. Vasoconstrictor drugs such as norepinephrine are used to:

- A. Increase heart rate only
- B. Decrease systemic vascular resistance
- C. Increase blood pressure and perfusion
- D. Decrease cardiac workload

Answer: C

23. Positive inotropic drugs (e.g., dobutamine) help by:

- A. Decreasing contractility
- B. Improving cardiac output
- C. Decreasing oxygen delivery
- D. Reducing venous return

Answer: B

24. Define shock and explain its main pathophysiological mechanism.

25. List the main types of shock and one cause of each.

26. Describe the four stages of shock.

27. Explain the compensatory mechanisms activated during the early stages of shock.

28. Describe the cellular changes during the progressive stage.

29. List five laboratory findings in shock and their significance.

30. Write the management goals in shock.

Case-Based / Application Questions

Case 1:

A 45-year-old man presents with severe vomiting and diarrhea for three days. BP: 85/50 mmHg, HR: 120 bpm, cold clammy skin, and urine output < 20 mL/hr.

Identify the type of shock.

Explain the underlying pathophysiology.

Outline the immediate management.

Answer:

Type: Hypovolemic shock

Pathophysiology: Loss of extracellular fluid → ↓ venous return → ↓ CO
→ ↓ tissue perfusion

Management: IV crystalloids, oxygen therapy, monitor urine output, treat cause (rehydration)

Case 2:

A 65-year-old diabetic woman with pneumonia becomes hypotensive (BP 80/45 mmHg), febrile, and tachycardic.

Identify the type of shock.

Explain why vasodilation occurs.

What is the first-line vasopressor in this condition?

Answer:

Type: Septic shock (distributive)

Vasodilation due to release of endotoxins and cytokines → loss of vascular tone

First-line vasopressor: Norepinephrine

Case 3:

A patient after a spinal cord injury develops hypotension and bradycardia. Skin is warm and dry.

What type of shock is this?

Explain the mechanism of bradycardia.

State the initial management.

Answer:

Type: Neurogenic shock

Mechanism: Loss of sympathetic tone → unopposed vagal stimulation
→ bradycardia

Management: Maintain airway, fluids, vasopressors, atropine for bradycardia

Case 4:

A patient develops sudden hypotension, dyspnea, and wheezing after receiving IV antibiotics.

What type of shock is this?

What is the drug of choice?

Mention two other supportive treatments.

Answer:

Type: Anaphylactic shock

Drug of choice: Epinephrine (IM/IV)

Supportive: Oxygen therapy, IV fluids, antihistamines, corticosteroids

Chest trauma questions

Q1 (MCQ): Which ribs are most commonly fractured in chest trauma?

- A) 1–3
- B) 5–9
- C) 10–12
- D) All ribs equally

Answer: B) 5–9

Rationale: The mid-thoracic ribs are most exposed and vulnerable to direct trauma.

Q2 (MCQ): Fractures of the first three ribs are dangerous because they may injure:

- A) Heart
- B) Subclavian artery
- C) Liver
- D) Diaphragm

Answer: B) Subclavian artery

Rationale: These ribs are protected by muscles and clavicle; fracturing them usually requires significant force, risking injury to major vessels.

Q3 (Short Answer): List common clinical manifestations of rib fractures.

Answer: Pain, tenderness, bruising, crepitus, muscle spasm, shallow breathing.

Q4 (True/False): Lower rib fractures can injure the spleen or liver.

Answer: True

Q5 (MCQ): Which nursing intervention is most important for rib fracture management?

- A) Strict bed rest
- B) Encourage deep breathing and coughing
- C) Avoid analgesics
- D) Apply rigid chest binders

Answer: B) Encourage deep breathing and coughing

Rationale: Prevents atelectasis and pneumonia; rigid binding is no longer recommended.

Q6 (MCQ): Flail chest is defined as:

- A) One rib fracture
- B) Three or more ribs fractured in two or more places
- C) Any rib fracture with lung contusion
- D) Rib fracture with hemothorax

Answer: B) Three or more ribs fractured in two or more places

Q7 (MCQ): Paradoxical chest wall motion is:

- A) Outward movement during inspiration
- B) Inward movement during inspiration
- C) Normal chest wall movement
- D) Chest expansion is equal

Answer: B) Inward movement during inspiration

Q8 (Short Answer): Name three complications of flail chest.

Answer: Pulmonary contusion, atelectasis, respiratory compromise, hypoxemia, pneumonia.

Q9 (MCQ): The most appropriate management of flail chest includes:

- A) Immediate surgery for all cases
- B) Pain control, oxygen therapy, mechanical ventilation if needed

- C) Only chest binders
- D) Restriction of movement

Answer: B) Pain control, oxygen therapy, mechanical ventilation if needed

Q10 (MCQ): Pneumothorax is:

- A) Blood in pleural space
- B) Air in pleural space
- C) Fluid in alveoli
- D) Collapse of trachea

Answer: B) Air in pleural space

Q11 (MCQ): Which of the following is NOT a type of pneumothorax?

- A) Simple
- B) Tension
- C) Hemothorax
- D) Open

Answer: C) Hemothorax

Q12 (MCQ): The hallmark sign of tension pneumothorax is:

- A) Bradycardia
- B) Mediastinal shift
- C) Hemoptysis
- D) Cough

Answer: B) Mediastinal shift

Q13 (Short Answer): Emergency management for tension pneumothorax?

Answer: Needle decompression (2nd intercostal space, midclavicular line) followed by chest tube insertion.

Q14 (True/False): Oxygen therapy alone is sufficient for a large tension pneumothorax.

Answer: False

Q15 (MCQ): Hemothorax is:

- A) Air in pleural space
- B) Blood in pleural space
- C) Infection in lung
- D) Pulmonary edema

Answer: B) Blood in pleural space

Q16 (MCQ): Massive hemothorax is defined as:

- A) >500 mL blood
- B) >1000 mL blood
- C) >1500 mL blood
- D) Any visible blood in pleura

Answer: C) >1500 mL blood

Q17 (Short Answer): Clinical signs of hemothorax?

Answer: Hypovolemic shock, decreased/absent breath sounds, dull percussion, dyspnea, distended neck veins.

Q18 (MCQ): Management of hemothorax includes:

- A) Only oxygen
- B) Chest tube drainage and fluid resuscitation
- C) Analgesics only
- D) Antibiotics only

Answer: B) Chest tube drainage and fluid resuscitation

Q19 (MCQ): Cardiac tamponade results in:

- A) Accumulation of air around heart
- B) Fluid accumulation in pericardial sac compressing the heart
- C) Myocardial infarction
- D) Pneumothorax

Answer: B) Fluid accumulation in pericardial sac compressing the heart

Q20 (MCQ): Beck's triad includes:

- A) Hypertension, bradycardia, muffled heart sounds
- B) Hypotension, jugular vein distension, muffled heart sounds
- C) Dyspnea, chest pain, cough
- D) Fever, hypotension, tachycardia

Answer: B) Hypotension, jugular vein distension, muffled heart sounds

Q21 (Short Answer): Emergency treatment for cardiac tamponade?

Answer: Pericardiocentesis to remove fluid

Q22 (MCQ): Key nursing intervention for chest trauma?

- A) Monitor vitals and SpO₂
- B) Avoid oxygen
- C) Restrict fluids
- D) Bed rest only

Answer: A) Monitor vitals and SpO₂

Q23 (MCQ): Why are opioids used cautiously in chest trauma?

- A) Cause hypotension only
- B) Cause respiratory depression, worsening hypoxemia
- C) Reduce pain completely
- D) Increase heart rate

Answer: B) Cause respiratory depression, worsening hypoxemia

Q24 (Short Answer): How should a patient with a chest tube be positioned?

Answer: Comfortable, upright if possible, avoid kinking the tube; turn/reposition every 1.5–2 hours.

Q25 (MCQ): Chest X-ray in chest trauma can detect:

- A) Rib fractures
- B) Pneumothorax/hemothorax
- C) Mediastinal shift
- D) All of the above

Answer: D) All of the above

Q26 (Short Answer): ABG findings in flail chest?

Answer: Hypoxemia, hypercapnia, respiratory acidosis

Q27 (MCQ): Why is ECG important in chest trauma?

- A) To detect fractures
- B) To detect dysrhythmias
- C) To check for hemothorax
- D) Not needed

Answer: B) To detect dysrhythmias

Code management questions

1. Which of the following best defines cardiac arrest?

- a) Slow heart rate
- b) Cessation of effective cardiac activity
- c) Chest pain only
- d) Shortness of breath

Answer: b) Cessation of effective cardiac activity

Explanation: Cardiac arrest occurs when the heart stops pumping effectively, leading to loss of circulation.

2. The first step in one-person CPR is to:

- a) Start chest compressions
- b) Check responsiveness and call for help
- c) Give 2 breaths
- d) Attach AED

Answer: b) Check responsiveness and call for help

Explanation: Safety and calling for help come before starting CPR.

3. The recommended sequence for CPR according to AHA guidelines is:

- a) ABC – Airway, Breathing, Compressions
- b) CAB – Compressions, Airway, Breathing
- c) BAC – Breathing, Airway, Compressions
- d) BCA – Breathing, Compressions, Airway

Answer: b) CAB – Compressions, Airway, Breathing

Explanation: AHA emphasizes starting with chest compressions (CAB).

4. What is the correct compression rate for adult CPR?

- a) 50–60/min
- b) 60–80/min
- c) 100–120/min
- d) 140–160/min

Answer: c) 100–120/min

Explanation: High-quality CPR requires 100–120 compressions per minute.

5. Which medication is indicated for pulseless ventricular tachycardia or ventricular fibrillation unresponsive to defibrillation?

- a) Atropine
- b) Epinephrine
- c) Amiodarone
- d) Lidocaine

Answer: c) Amiodarone

Explanation: Amiodarone is used after shock-resistant VF/pulseless VT.

6. What is the primary role of the code team leader?

- a) Perform chest compressions only
- b) Document the event
- c) Direct and coordinate the resuscitation efforts
- d) Administer medications

Answer: c) Direct and coordinate the resuscitation efforts

7. Which of the following statements about defibrillation is true?

- a) It is used for asystole
- b) It delivers an electrical shock to terminate arrhythmias
- c) It is only used in children
- d) It must be repeated every 30 seconds regardless of rhythm

Answer: b) It delivers an electrical shock to terminate arrhythmias

8. During CPR, the chest should be compressed to a depth of:

- a) 1 inch (2.5 cm)
- b) 2 inches (5 cm)
- c) 3 inches (7.5 cm)
- d) 4 inches (10 cm)

Answer: b) 2 inches (5 cm) for adults

9. Which of the following is a key goal of post-resuscitation care?

- a) Maintain airway and oxygenation
- b) Monitor cardiac rhythm
- c) Prevent neurological injury

d) All of the above

Answer: d) All of the above

10. **Therapeutic hypothermia after cardiac arrest is used to:**

a) Stop heart function temporarily

b) Reduce neurological injury

c) Increase blood pressure

d) Replace CPR

Answer: b) Reduce neurological injury

11. **Which of the following rhythms is shockable by a defibrillator?**

a) Asystole

b) Pulseless electrical activity

c) Ventricular fibrillation

d) Bradycardia

Answer: c) Ventricular fibrillation

12. **How many ventilations should be given after 30 compressions in adult CPR?**

a) 1

b) 2

c) 3

d) 5

Answer: b) 2

13. **What is the recommended hand position for adult chest compressions?**

a) Center of the chest, lower half of the sternum

b) Left side of the chest

c) Right side of the chest

d) Upper chest near the clavicle

Answer: a) Center of the chest, lower half of the sternum

14. Which of the following is an early sign of cardiac arrest?

- a) Chest pain
- b) Sudden collapse
- c) Fever
- d) Nausea

Answer: b) Sudden collapse

15. What is the recommended compression-to-ventilation ratio for two-person CPR in adults?

- a) 15:2
- b) 30:2
- c) 50:2
- d) 100:2

Answer: b) 30:2

16. Which device provides continuous chest compressions without fatigue?

- a) AED
- b) Manual defibrillator
- c) Mechanical chest compression device (e.g., LUCAS)
- d) Oxygen bag

Answer: c) Mechanical chest compression device

17. Which of the following is NOT part of the primary survey in CPR?

- a) Airway assessment
- b) Breathing assessment
- c) Circulation assessment
- d) Full neurological exam

Answer: d) Full neurological exam

18. What is the recommended depth of chest compressions in children?

- a) 1 inch (2.5 cm)
- b) 1.5 inches (4 cm)
- c) 2 inches (5 cm)
- d) 3 inches (7.5 cm)

Answer: b) 1.5 inches (4 cm)

19. **Which of the following drugs is first-line for asystole?**

- a) Epinephrine
- b) Amiodarone
- c) Lidocaine
- d) Atropine

Answer: a) Epinephrine

20. **During a code, who is responsible for recording all interventions and times?**

- a) Code team leader
- b) CPR provider
- c) Recorder/scribe
- d) Pharmacist

Answer: c) Recorder/scribe

True/False Questions

21. **CPR should be initiated within 4–6 minutes of cardiac arrest.**

Answer: True

22. **The compression-to-breath ratio for adult one-person CPR is 30:2.**

Answer: True

23. **Atropine is indicated for asystole as the first-line drug.**
Answer: False (Epinephrine is first-line)
24. **Automated external defibrillators (AEDs) can analyze cardiac rhythm and advise whether a shock is needed.**
Answer: True
25. **Endotracheal intubation is considered a BLS measure.**
Answer: False (it is ALS)
26. **High-quality CPR requires minimal interruptions in chest compressions.**
Answer: True
27. **Children with cardiac arrest usually present with a primary cardiac cause.**
Answer: False (often respiratory causes)
28. **Epinephrine is contraindicated in ventricular fibrillation.**
Answer: False
29. **Chest compressions should be paused only for defibrillation or rhythm check.**
Answer: True
30. **CPR is ineffective if performed on a firm surface.**
Answer: False (firm surface is recommended)
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Short Answer Questions

31. **Define cardiac arrest and explain why immediate intervention is necessary.**
Answer: Cardiac arrest is the sudden cessation of effective cardiac

activity. Immediate intervention is necessary to restore circulation and prevent brain injury or death.

32. List at least 5 causes of cardiopulmonary arrest.

Answer: Myocardial infarction, arrhythmias, respiratory failure, drowning, trauma.

33. Explain the roles of at least three members of the code team.

Answer: Leader – directs resuscitation; Compressor – performs chest compressions; Recorder – documents events; Airway provider – manages airway; IV/medication provider – administers drugs.

34. Describe the steps to establish an airway and provide ventilations during CPR.

Answer: Open airway using head-tilt/chin-lift or jaw-thrust, provide 2 rescue breaths after 30 compressions, ensure chest rises.

35. Discuss the rationale for therapeutic hypothermia in post-cardiac arrest patients.

Answer: Cooling reduces metabolic demand and protects the brain from hypoxic injury.

36. What are the main differences between defibrillation and cardioversion?

Answer: Defibrillation is unsynchronized and used in VF/pulseless VT; cardioversion is synchronized and used for unstable arrhythmias with a pulse.

37. Name three medications used during a code and their indications.

Answer: Epinephrine – asystole/PEA; Amiodarone – VF/pulseless VT; Atropine – bradycardia.

38. **Explain the importance of using a backboard during CPR.**

Answer: Provides a firm surface for effective chest compressions.

39. **Describe how to assess responsiveness in an unresponsive patient.**

Answer: Shake the patient gently and shout, “Are you okay?”

40. **List the signs of effective chest compressions during CPR.**

Answer: Chest rises and falls, palpable carotid pulse returns, improved skin color.

41. **Explain the steps to operate an AED.**

Answer: Turn on AED, attach pads, allow AED to analyze rhythm, deliver shock if advised, resume CPR.

42. **What precautions should be taken when defibrillating a patient with an implanted pacemaker?**

Answer: Place pads at least 1 inch away from pacemaker, follow standard shock protocol.

43. **Describe the post-resuscitation monitoring process.**

Answer: Monitor airway, breathing, circulation, cardiac rhythm, neurological status, blood pressure, and oxygen saturation.

44. **Explain the significance of early defibrillation in ventricular fibrillation.**

Answer: Early defibrillation increases the chance of restoring normal rhythm and survival.

45. **Describe the difference in CPR technique between adults, children, and infants.**

Answer: Adults – 2 hands, depth 2 inches; Children – 1–2 hands, depth 1.5 inches; Infants – 2 fingers, depth 1.5 inches.

46. **Discuss the role of oxygen therapy during and after CPR.**

Answer: Provides oxygen to vital organs and prevents hypoxia during resuscitation.

47. **Explain how to manage a choking adult patient before cardiac arrest occurs.**

Answer: Perform Heimlich maneuver (abdominal thrusts) until obstruction is cleared.

48. **Describe the steps to convert an irregular heart rhythm using synchronized cardioversion.**

Answer: Assess rhythm, apply pads, set energy level, synchronize with R wave, deliver shock while monitoring patient.

49. **List complications that can occur during CPR and how to minimize them.**

Answer: Rib fractures, pneumothorax, aspiration; minimize by proper technique and careful handling.

50. **Explain the importance of debriefing and documentation after a code.**

Answer: Debriefing improves team performance, documentation ensures accurate record, legal and quality assurance purposes.

Respiratory failure questions

Q1: Respiratory failure is defined as:

- A) Inability to produce surfactant
- B) Inability to maintain adequate gas exchange
- C) Chronic lung inflammation
- D) Excessive alveolar ventilation

Answer: B

Q2: Type I respiratory failure is characterized by:

- A) Low PaO₂ and high PaCO₂
- B) Low PaO₂ and normal PaCO₂
- C) High PaO₂ and low PaCO₂
- D) High PaO₂ and high PaCO₂

Answer: B

Q3: Type II respiratory failure is also called:

- A) Oxygenation failure
- B) Hypercapnic ventilatory failure
- C) Alveolar shunting
- D) Perfusion failure

Answer: B

Q4: A patient with PaO₂ 55 mmHg, PaCO₂ 52 mmHg, and pH 7.28 has:

- A) Type I respiratory failure
- B) Type II respiratory failure
- C) Normal ABG
- D) Metabolic alkalosis

Answer: B

Q5: Which of the following is an extra-pulmonary cause of respiratory failure?

- A) Pneumonia
- B) Pulmonary embolism
- C) Myasthenia gravis
- D) ARDS

Answer: C

Q6: Which of the following is an intra-pulmonary cause?

- A) Spinal cord trauma

- B) COPD
- C) Guillain-Barre syndrome
- D) Obesity

Answer: B

Q7: Pulmonary embolism primarily affects:

- A) Upper airways
- B) Pulmonary circulation
- C) Neuromuscular system
- D) Pleura

Answer: B

Q8: Massive obesity may lead to respiratory failure by:

- A) Alveolar-capillary shunting
- B) Upper airway obstruction
- C) Decreased thoracic compliance
- D) Pulmonary edema

Answer: C

Q9: Which drug overdose is commonly associated with extra-pulmonary respiratory failure?

- A) Antibiotics
- B) Opioids
- C) Antihistamines
- D) Diuretics

Answer: B

Q10: The hallmark of acute respiratory failure is:

- A) Hypercapnia
- B) Hypoxemia
- C) Respiratory alkalosis

D) Pulmonary hypertension

Answer: B

Q11: Alveolar hypoventilation leads to:

A) Hypoxemia with hypercapnia

B) Hypoxemia with normal PaCO_2

C) Hypercapnia only

D) Alkalosis

Answer: A

Q12: A V/Q mismatch occurs when:

A) Ventilation exceeds perfusion or vice versa

B) Alveoli collapse completely

C) Oxygen dissolves in plasma

D) CO_2 is adequately removed

Answer: A

Q13: Intrapulmonary shunt is defined as:

A) Blood bypassing non-ventilated alveoli

B) Reduced oxygen demand

C) Increased cardiac output

D) Alveolar hyperventilation

Answer: A

Q14: Pulmonary edema contributes to hypoxemia primarily by:

A) Increasing cardiac output

B) Flooding alveoli, causing shunting

C) Reducing respiratory rate

D) Increasing V/Q ratio

Answer: B

Q15: Which of the following is a CNS manifestation of respiratory failure?

- A) Bradycardia
- B) Restlessness
- C) Edema
- D) Cyanosis

Answer: B

Q16: Pulmonary signs of respiratory failure include all except:

- A) Dyspnea
- B) Tachypnea
- C) Use of accessory muscles
- D) Bradycardia

Answer: D

Q17: Skin changes in respiratory failure include:

- A) Jaundice
- B) Cyanosis and pallor
- C) Petechiae
- D) Erythema

Answer: B

Q18: A patient with respiratory acidosis may show:

- A) Hyperventilation
- B) Tachypnea initially
- C) CNS depression
- D) All of the above

Answer: D

Q19: Which diagnostic test is gold standard for confirming respiratory failure?

- A) Chest X-ray
- B) Arterial Blood Gas (ABG)
- C) Pulmonary function test
- D) Bronchoscopy

Answer: B

Q20: Which ABG result indicates Type II respiratory failure?

- A) PaO₂ 60 mmHg, PaCO₂ 40 mmHg
- B) PaO₂ 55 mmHg, PaCO₂ 52 mmHg
- C) PaO₂ 80 mmHg, PaCO₂ 35 mmHg
- D) PaO₂ 90 mmHg, PaCO₂ 30 mmHg

Answer: B

Q21: Bronchoscopy in respiratory failure is used for:

- A) Measuring PaO₂
- B) Airway surveillance and specimen retrieval
- C) Nutrition support
- D) Ventilation

Answer: B

Q22: Thoracic CT scan is useful to:

- A) Determine electrolyte levels
- B) Detect underlying structural lung disease
- C) Improve oxygenation
- D) Measure PaCO₂

Answer: B

Q23: The first priority in acute respiratory failure is:

- A) Nutrition support
- B) Correcting underlying cause
- C) Administering diuretics

D) Physical therapy

Answer: B

Q24: Oxygen therapy should maintain SpO₂ at:

A) 70–80%

B) 88–95%

C) 60–75%

D) 100% at all times

Answer: B

Q25: Non-invasive ventilation (NIV) is indicated for:

A) All patients with Type II failure

B) Patients with mild to moderate hypoxemia without severe acidosis

C) Patients needing surgery

D) Only patients with pneumonia

Answer: B

Q26: Bronchodilators help by:

A) Reducing inflammation

B) Relaxing smooth muscles to improve ventilation

C) Increasing cardiac output

D) Correcting acidosis

Answer: B

Q27: Corticosteroids are used to:

A) Reduce airway inflammation

B) Increase CO₂ retention

C) Promote sedation

D) Improve cardiac output

Answer: A

Q28: Sodium bicarbonate is indicated in respiratory failure when:

A) pH <7.10 and refractory to other therapy

- B) pH >7.45
- C) PaCO₂ <40 mmHg
- D) SpO₂ >95%

Answer: A

Q29: Patient positioning in unilateral lung disease should be:

- A) Healthy lung in dependent position
- B) Diseased lung in dependent position
- C) Supine at all times
- D) Prone only

Answer: A

Q30: Frequent repositioning helps:

- A) Prevent hypoxemia and improve V/Q matching
- B) Cause fatigue
- C) Increase cardiac workload
- D) Reduce secretions

Answer: A

Q31: How often should incentive spirometry be performed?

- A) Once a day
- B) At least 10 deep breaths per hour
- C) Only when patient is hypoxic
- D) Every 4–6 hours

Answer: B

Q32: Deep breathing exercises are recommended to:

- A) Reduce oxygen consumption
- B) Prevent atelectasis
- C) Reduce pulmonary perfusion

D) Increase heart rate

Answer: B

Q33: Coughing should be avoided unless:

A) Patient has fever

B) Secretions are present

C) SpO₂ >95%

D) Patient is on oxygen

Answer: B

Q34: Patient education should include all except:

A) Breathing techniques

B) Smoking cessation

C) Energy conservation

D) Only medications without lifestyle changes

Answer: D

Q35: Signs of pulmonary infection the patient should monitor include:

A) Sputum color change, fever, shortness of breath

B) Headache only

C) Abdominal pain

D) Palpitations

Answer: A

Q36: Ischemic anoxic encephalopathy is caused by:

A) Hypercapnia, hypoxemia, acidosis

B) Hypotension only

C) Infection

D) Electrolyte imbalance

Answer: A

Q37: Venous thromboembolism can be prevented by:

- A) Immobility
- B) Low molecular weight heparin and pneumatic compression
- C) Diuretics
- D) High salt diet

Answer: B

Q38: Dysrhythmias in respiratory failure are commonly caused by:

- A) Hyperoxemia
- B) Hypoxemia, acidosis, electrolyte imbalance
- C) Diuresis
- D) Low blood pressure only

Answer: B

Q39: Gastrointestinal bleeding risk can be reduced by:

- A) Histamine receptor antagonists or proton pump inhibitors
- B) NSAIDs
- C) High protein diet
- D) Mechanical ventilation

Answer: A

Q40: Patients should be taught to use which breathing technique to improve ventilation?

- A) Pursed-lip breathing
- B) Shallow chest breathing
- C) Breath-holding only
- D) Rapid hyperventilation

Answer: A

Q41: Energy conservation techniques help by:

- A) Increasing oxygen demand

- B) Reducing fatigue and oxygen consumption
- C) Preventing infection
- D) Decreasing PaCO₂ only

Answer: B

Q42: Pulmonary rehabilitation includes:

- A) Exercise, breathing techniques, and patient education
- B) Surgery only
- C) Diet changes only
- D) Bed rest only

Answer: A

Q43: When should enteral nutrition be initiated for malnourished patients on mechanical ventilation?

- A) Within 24 hours
- B) After 3 days
- C) Only after extubation
- D) Not required

Answer: A

Q44: What is the goal of oxygen therapy in respiratory failure?

- A) PaO₂ 55–80 mmHg, SpO₂ 88–95%
- B) PaO₂ <50 mmHg
- C) SpO₂ <80%
- D) PaO₂ >100 mmHg always

Answer: A

Q45: Patients with alveolar hypoventilation should be positioned:

- A) Semirecumbent or sitting
- B) Supine
- C) Lying on diseased lung
- D) Prone only

Answer: A

Q46: Which electrolyte imbalance commonly contributes to dysrhythmias in respiratory failure?

- A) Hyperkalemia or hypokalemia
- B) Hypernatremia only
- C) Hypocalcemia only
- D) Hypermagnesemia only

Answer: A

Q47: Mechanical ventilation tidal volume recommended is:

- A) 6 mL/kg
- B) 10 mL/kg
- C) 12 mL/kg
- D) 15 mL/kg

Answer: A

Q48: Frequent monitoring with pulse oximetry helps to:

- A) Warn of desaturation early
- B) Replace ABG analysis
- C) Prevent infection
- D) Increase tidal volume

Answer: A

Q49: Neuromuscular blocking agents may be used to:

- A) Facilitate ventilation and decrease oxygen consumption
- B) Reduce fever
- C) Treat hypoxemia
- D) Increase heart rate

Answer: A

Q50: The most important preventive intervention for reoccurrence of respiratory failure is:

- A) Smoking cessation and management of precipitating factors
- B) Surgery
- C) Bed rest
- D) High-flow oxygen at home

Answer: A

question bank critical care nursing

Pain mangment

1-Types of pain affect nerve ending of the organs include which of the following

- a-Acute pain
- b-Chronic pain
- c-Neuropathic pain
- d- Nociceptive pain

2- Chronic pain characterized by

- a- Duration less than 6 months
- b- Occur with acute onset
- c- Duration more than 6 months
- d- Need rapid management

3- Neuropathic Pain characterized by

- a- Occur with acute onset
- b- Arises from a lesion or disease affecting CNS
- c- Duration more than 6 months
- d- Involves organs such as the heart, stomach, and liver.

4- Behavior Pain Scale (BPS) assess which of the following

- a- Facial Expression
- b- Upper Limbs
- c- Compliance with Ventilation
- d- All of the above

5- Grimacing in the facial expression scale indicate

- a- Indicate patient in rest
- b- Indicate patient in moderate pain
- c- Indicate patient in sever pain
- d- Non of the above

6-Sighing, moaning in Vocalization (non intubated patients)

Indicate which of the following

- a-Patient is relaxed
- b- Patient in mild pain
- c-Patient in moderate pain
- d-patient in sever pain

List four point of the following

1- Types of pain

2- Non pharmacological treatment of pain

ECG

1- PR intervals means which of the following

- A- Distance between atrial contraction and ventricular relaxation
- B- Distance between atrial contraction and ventricular contraction
- C- Distance between atrial relaxation and ventricular contraction
- D- Distance between atrial relaxation and ventricular relaxation

2- T wave appear in ECG in which phase

- A- Atrial depolarization
- B- Ventricular depolarization
- C- Atrial repolarization
- D- Ventricular repolarization

3- U wave appear in ECG at deficiency of which of the following

- A- Na
- B- K

C- Ca

D- Mg

4- Normal sinus rhythm characterized by

A- Present of atrial and ventricular depolarization and repolarization

B- Present of atrial depolarization only

C- Present of ventricular depolarization

D- Present of atrial depolarization and ventricular re-polarization

5- Calculate the heart rate for the following



Hr = 75 b/m



Hr = $1500/39=38$ b/m



Hr =150b/m

6- During cardiac depolarisation which of the following electrolyte enter the heart muscle

- A- Na
- B- K
- C- Ca
- D- Ca and Na

7- Which of the following site is responsible for initiating atrial electrical impulses?

- A- Bundle of His
- B- Purkinje fibers
- C- Sinoatrial (SA) node

D- Atrioventricular (AV) node

8- Atrial depolarization indicates which of the following on an electrocardiogram rhythm strip?

A- U waves i

B- QRS complex i

C- T waves i

D- P waves

i

Cardiac dysrthmia

MCQ

- 1- Sinus tachycardia characterized by which of the following**
 - a- Rapid sinus rhythm**
 - b- Rapid irregular rhythm**
 - c- Slow regular rhythm**
 - d- Regular sinus rhythm**
- 2- Causes of sinus bradycardia includes which of the following**
 - a- Stress**
 - b- Hyperkalemia**
 - c- Pericarditis**
 - d- Atropine administration**
- 3- Atrial fibrillation characterized by which of the following**

- a- Rapid sinus rhythm
 - b- Slow irregular rhythm
 - c- Rapid irregular rhythm with absence of p wave
 - d- Slow regular rhythm
- 4- Causes of AF includes which of the following
- a- Hypertension
 - b- Valvular heart disease
 - c- Coronary heart disease
 - d- All of the above
- 5- Acute management of AF include which of the following
- a- Atropine and adrenaline
 - b- Cardio version
 - c- Defibrillation
 - d- Pacemaker insertion
- 6- Premature ventricular contraction characterized by
- a- Rapid regular rhythm
 - b- Rapid irregular rhythm
 - c- Absence of p wave
 - d- Wide QRS complex

7- Causes of ventricular tachycardia include

- a- Congestive Heart failure
- b- Heart block
- c- Hyperkalemia
- d- Hypocalcemia

8- Ventricular fibrelation characterized by

- a- Rapid sinus rhythm
- b- Rapid irregular rhythm
- c- Irregular rhythm with absence of QRS
- d- Slow irregular rhythm

9- Management of VF includes

- a- Cardio version

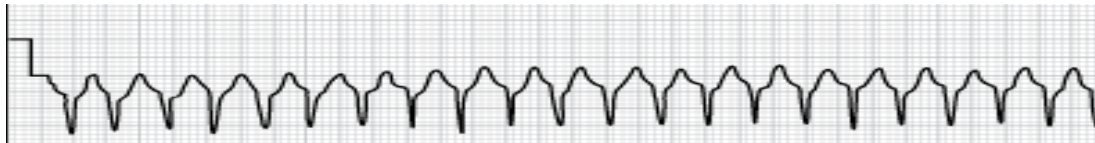
- b- Defibrillation
- c- Rate control medication
- d- Anticoagulant medication

10-Mention type of arrhythmia to the following strip



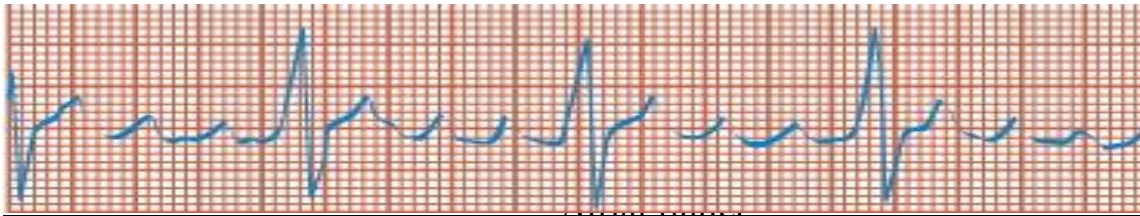
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Sinus tachycardia



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Ventricular tachycardia



Acute Coronary Syndrom

MCQ

1- Causes of unatatable angina includes

- a- Heart failure
- b- Coronary atherosclerosis
- c- Coronary artery thrombosis
- d- Aneurysm

2- Signs and symptoms of MI include which of the following

- a- Moderate chest pain releved by sedation

b- Sever chest pain not releved by sedation

c- Abdominal pain

d- Fever

3- Complications of MI include which of the following

a- Heat block

b- Ventricular tachycardia

c- Ventricular aneurysm

d- Hypertension

4- Acute treatment of MI include which of the following

a- Cardiovesion

b- Defibrillation

c- Thrombolytic therapy

d- Adenosine

List four point of the following

1- Common Nursing diagnosis for patients with myocardial infarction

2- Causes of myocardial infarction

3- Complications of myocardial infarction

4- Medical manegment of patients with MI

Cardiac surgery

MCQ

- 1- Complication of cardiac surgery
 - a- Hypertension
 - b- Aneurysm
 - c- Cardiac tamonade
 - d- Pneumothorax
- 2- Indication of cardiac surgery include
 - a- Multivessles coronary artery disease
 - b- Unstable angina
 - c- Heat block
 - d- Hypertension
- 3- Types of open heart surgery includes all of the following except
 - A- Coronary artery bypass graft
 - B- Valvular heat disease
 - C- Internal applation
 - D- Heat trasplantation

Give short account on the following :

- 1- Heart lunge machine

- 2- Prepration of patients preoperative phase

- 3- Complication post open heat surgery

4- Nursing diagnosis for patients post open heart surgery